Animesh Jayant

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Placement Director:	Matthias Doepke m.doepke@lse.ac.uk
Graduate Administrator:	Emma Taverner e.taverner@lse.ac.uk

Doctoral Studies

London School of Economics
PhD Candidate in Economics — Expected Completion 2027
MRes Economics, *Distinction* — 2022

REFERENCES

Rocco Macchiavello
Department of Management
London School of Economics
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Robin Burgess
Department of Economics
London School of Economics
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Oriana Bandiera
Department of Economics
London School of Economics
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Daniel Sturm
Department of Economics
London School of Economics
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Clare Balboni
Department of Economics
London School of Economics
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Prior Education

University of Chicago — 2019
MA Research Methods

Indian Institute of Technology Bombay — 2013
BTech Metallurgical Engineering & Materials Science

Fields

Primary: Development Economics
Secondary: Industrial Organization, Organizational Economics

Job Market Paper

Beyond Yields: How Climate Change Affects Farmers Through Market Structure

Abstract: Climate change hurts farmers by lowering crop yields. In this paper, I show that it also hurts farmers by changing the structure of the markets they sell in. To document this, I exploit extreme drought shocks in the water-intensive Indian sugar industry where 8M smallholders sell perishable cane to oligopsonistic sugar mill intermediaries. I assemble a newly digitized 35-year administrative panel on the universe of Indian sugar mills, and a georeferenced farmer-mill transaction panel on 3M cane farmers. Using a staggered event study design, I show that droughts concentrate markets by removing the least productive mills, and trigger long-term productivity growth within survivors through management upgrading. Yet farm-gate cane prices fall as these gains are captured by survivors through increased buyer power. Reduced cane prices account for one half of farmers' long-term income losses - on par with yield losses themselves - pointing to a large role for market structure in shaping farmers' climate burden. To quantify aggregate welfare impacts, I build and estimate an equilibrium model of the Indian sugar sector that embeds a management upgrading choice for sugar mills. Drought-induced market restructuring raises aggregate mill productivity by 12 percent through both selection and within-firm channels, but long-term farmer welfare declines by 8 percent as markdowns widen.

Working Papers

1. **Captive Markets: How Zoning Hinders Productivity Growth in Agriculture**

   Abstract: Zoning regulations that guarantee farmers a buyer by partitioning markets into exclusive catchments are pervasive in low-income agriculture, yet their consequences for productivity remain largely unknown. In this paper, I show that such regulations suppress productivity growth and welfare on both sides of the farm gate, using the Indian sugar industry as a laboratory. I exploit Maharashtra's 1997 abolition of cane reservation areas, which tied each farmer and sugar mill to an exclusive zone, as a natural experiment that introduced competition into cane procurement. To overcome measurement challenges that have hindered empirical progress on this question, I draw on novel long-term administrative records covering the universe of Indian sugar mills, with plant-level measures of technology, management, and production. Using an event-study design, I find that removing zoning raised within-mill sugar recovery – a direct measure of processing productivity – by 7 percent, driven entirely by management upgrading. Cane prices paid to farmers also increased by 5 percent. To quantify aggregate effects, I estimate a structural model of the sector and find that abolishing cane zoning nationwide would have raised total welfare by 8 percent. The results show that a long-standing regulatory norm in agriculture can be a first-order barrier to productivity growth.

2. **The Interaction of Contracting Frictions and Market Power in Agricultural Supply Chains** (with Rocco Macchiavello and Juan Quicana)

   Abstract: Agricultural value chains in developing countries connect a large number of smallholders to a small set of downstream buyers under weak contract enforcement. This review examines how two important features of these markets – buyer power and relational contracting – jointly shape prices, production, investment, and farmer welfare. We organize recent empirical evidence around a common framework linking farmers, intermediaries, and downstream buyers. Three main conclusions emerge. First, buyer power depends not only on the number of buyers but also on their conduct and farmers'

ability to switch across buyers and activities. Second, market power at different stages of the value chain can interact in ways that make the effects of policy difficult to predict. Intermediaries may exercise market power while also providing valuable services, so higher farm-gate prices do not necessarily imply higher farmer welfare. Third, market power and relational contracting are closely connected: competition improves outside options but can also reduce the rents needed to sustain informal agreements. We argue that future work should study market power and relational contracting jointly and evaluate agricultural policies beyond their effects on prices alone.

Work in Progress

1. **Contract Design in Agroforestry Carbon Markets: Improving Performance and Building Climate Resilience** (with Shruti Bhimsaria)

   Abstract: Voluntary carbon markets can mobilize climate finance for low-income countries through nature-based solutions like agroforestry, which combines carbon sequestration with farmer resilience. Yet existing projects exhibit very low tree survival rates, reflecting two key failures in standard contracting models: adverse selection – as low-commitment landowners self-select into programs, and moral hazard – as farmer distrust of developers depresses maintenance effort. We study how contract design can mitigate these information asymmetries to improve market performance and strengthen climate adaptation. Partnering with a large agroforestry project developer in India, we implement a 2×2 cross-randomized controlled trial across 250 village clusters and 2,500 farmers. Each dimension of the factorial design targets a distinct information asymmetry: one varies contract features that screen on farmer commitment to address adverse selection, while the other varies developer-side investments that build relational trust to address moral hazard. We evaluate impacts on four outcome families: (a) selection; (b) tree survival and carbon sequestration measured via drone-based digital MRV; (c) income diversification and consumption smoothing under climate shocks; (d) and environmental co-benefits including soil health and groundwater recharge.

2. **Optimal Membership in the Cooperative Firm: Evidence from Indian Sugar Cooperatives**

   Abstract: Cooperative firms must decide how many of their input suppliers to admit as members. More members expand the captive supply base and increase processing throughput, but dilute per-member surplus through profit-sharing and extension service provision. Despite cooperatives' prevalence in agricultural markets worldwide, little is known about how this trade-off is resolved in practice. I study this question in India's sugar industry, where cooperative mills procure sugarcane from both member and non-member farmers – yet only a third of suppliers hold membership. Using farmer- and mill-level panel data covering the universe of both cane farmers and sugar mills in Maharashtra, I develop and estimate a structural model in which cooperatives endogenously choose membership thresholds while competing with private mills for farmer supply in a spatial oligopsony. The model features a nested discrete choice framework for farmer mill selection, an oligopsony pricing game among sugar mills, and a cooperative membership rule that trades off supply expansion against dividend dilution and governance costs. Counterfactual simulations quantify the welfare consequences of mandating open membership, eliminating cooperatives, and adjusting membership policy under climate stress.

3. **Farm Size Matters in a Warming World: Evidence from Indian Agriculture**

   Abstract: Recent research highlights how climate damages can be mitigated through sectoral reallocation, crop reallocation, and reducing trade and migration barriers. In this project, I evaluate the potential for reallocation between agricultural and non-agricultural production on the basis of farm size to serve as a margin of climate adaptation, and examine how input market and sectoral mobility frictions can hamper this process. I study this question in the context of India, a highly climate-vulnerable economy comprising 25% of the world's farms, proceeding in three steps. First, I present empirical estimates showing that productivity losses from rising temperatures are only half as large on larger farms relative to smaller ones – implying substantial gains if climate-exposed countries could promote farm consolidation and facilitate labour reallocation from vulnerable small farms into non-agriculture. Second, I document that long-run responses to rising temperatures see a shrinking of average farm sizes, driven by a decline in the number of large farms rather than the consolidation that efficient adaptation would predict. This points to an important role for input market frictions. Third, to rationalize these findings, I develop a theoretical framework embedding input market and sectoral mobility frictions in a Melitz model, and use it to quantify how these frictions dampen the gains from climate adaptation.

Teaching

Guest Lecturer: LSE EC551 PhD Industrial Economics 2025

Lecturer: LSE Summer School MG206 Undergraduate Business Strategy 2026

LSE EC1C1: Undergraduate Econometrics 2021-2026

- Course Manager and TA to Professors Michael Callen and Michael Gmeiner
- LSE Teaching Award 2022–23: top 10% (∼8 out of 80 Department TAs)
- Average Teaching Evaluation: 4.8/5

LSE EC1A5: Undergraduate Microeconomics 2021-2026

- Course Manager and TA to Professor Alan Manning
- LSE Teaching Award 2022–23: top 10% (∼8 out of 80 Department TAs)
- Average Teaching Evaluation: 4.6/5

LSE Summer School MG206: Undergraduate Business Strategy 2023-2024

- Course Manager and TA to Professor Rocco Macchiavello
- Average Teaching Evaluation: 4.7/5

Research Grants

"Beyond Yields: How Climate Change Affects Farmers Through Market Structure." PEDL PhD Research Grant. £18,000

"Beyond Yields: How Climate Change Affects Farmers Through Market Structure." STICERD PhD Research Grant. £30,000

"Farm Size Matters in a Warming World: Evidence from Indian Agriculture". IGC Small Projects Facility. £20,000

"Captive Markets: How Zoning Hinders Productivity Growth in Agriculture", with Rocco Macchiavello. STICERD Staff Research Grant. £10,000

Presentations (* scheduled)

2026	Northeastern Universities Development Consortium (NEUDC)*	Cornell
	PEDL Conference on Climate & Environment at LSE EEE Week	LSE
	Paris Environmental Economics Conference	PSE
	Conference on Strategies for Resilient Entrepreneurship	Colombo

Professional Activities

Refereeing: Journal of Law, Economics and Organization, Review of Industrial Organization

Organizing Committee: LSE Environment Week (2023, 2024), LSE Environment Camp (2023, 2024, 2025)

Affiliations: LSE Economics of Environment and Energy Programme (EEE); Center for Economic Performance (CEP)

Honors, Scholarships, and Fellowships

LSE PhD Studentship	*2020-26*
LSE Economics Department Award	*2020-26*
University of Chicago DRW Fellowship	*2018-19*
University of Chicago Merit Fellowship	*2018-19*

Work Experience

2019-21	Research Assistant to Professor Eyal Frank, University of Chicago
2018-20	Research Assistant to Professor Fiona Burlig and Professor Anant Sudarshan, University of Chicago
2014-18	Research Assistant to Professor Michael Greenstone, Professor Robin Burgess, Professor Nicholas Ryan and Professor Anant Sudarshan, J-PAL India
2013-14	Management Consultant, PricewaterhouseCoopers India

Languages

English (Fluent), Hindi (Native), Marathi (Intermediate)

Citizenship

Indian

Last Updated: November 2026